ETH Zürich
Department of Computer Science

Paper Presentation: “Optimistic Active Exploration of Dynamical Systems”

Based on the work of Sukhija et al. (NeurIPS 2023)

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Foundations of Reinforcement Learning (263-5255-00L), Spring 2025

Outline

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Motivation: Beyond Task-Specific Models

- **Standard MBRL Goal:** Learn dynamics model f to solve a *specific* task (maximize reward R).
- **Problem:** Exploration driven by task reward R leads to biased models.
  - Agent only learns dynamics well in task-relevant state-action regions.
  - Model fails to generalize to *new* tasks requiring different parts of the state space.
- **Need:** Methods to learn *globally accurate* models, useful for *any* downstream task.
- **Idea:** Decouple exploration from extrinsic rewards. Explore to gain *information* about the system itself.

Goal of Task-Agnostic Exploration

Interact with the environment to build a dynamics model $\hat{f}$ that is accurate everywhere, enabling *zero-shot* (or few-shot) generalization to new tasks.

Introducing OPAX

- **OPAX:** Optimistic Active Exploration of Dynamical Systems.[1]
- **Core Idea:** Frame exploration as Bayesian optimal experimental design.
  - Actively seek interactions (trajectories) that maximally reduce *epistemic uncertainty* about the unknown dynamics f^*.
- **Key Components (Overview):**
  1. **Probabilistic Model:** Quantifies belief and uncertainty about f^* (e.g., Gaussian Process, Ensemble).
  2. **Information Gain Objective:** Formalizes uncertainty reduction as the goal.
  3. **Optimistic Planning:** Plans trajectories assuming the "most informative" plausible dynamics.

Promise of OPAX

Learn general-purpose world models efficiently, enabling robust planning and control across diverse tasks without task-specific retraining.

[1] **sukhija2023optimistic**.

Outline

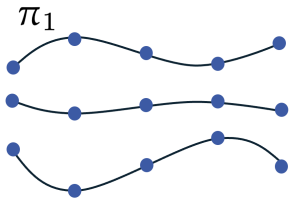
Problem Formulation (1/2)

Given an unknown discrete-time dynamical system $f^\star : \mathcal{S} \times \mathcal{A} \to \mathcal{S} \subseteq \mathbb{R}^d$:

$$s_{t+1} = f^\star(s_t, a_t) + w_t, \quad w_t \sim \mathcal{N}(0, I).$$

Goal: Design algorithms for learning $\hat{f} \approx f^\star$ that is good for any reward functions in episodic setting:

$$\mathcal{D} = \left\{ \begin{array}{c} \pi_1 \\ \\ \\ \pi_n \end{array} \right\}$$



Introduction video by Bhavya Sukhija. NeurIPS 2023.

Problem Formulation (2/2)

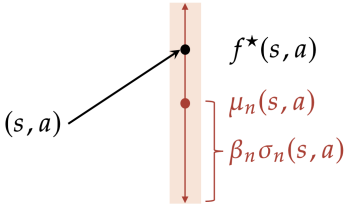
Our problem can be divided into two parts:

- Exploration (our focus!): Collecting data by executing policies.
- Estimation: Building model from data.

Well-Calibration Assumption

Given n data, we can construct confidence intervals:

$$|f_i^\star(s,a) - \mu_{n,i}(s,a)| \leqslant \beta_n \sigma_{n,i}(s,a) \quad \forall i \in [d], \text{ w.h.p.},$$



Outline

Related Work: Active Learning

	Bandit Optimization	Our Problem
Goal	$\max_x f^\star(x)$	Find $\hat{f} \approx f^\star$
$f^\star$	Objective	Dynamics
State space	No	Continuous
Planning	No	Yes
Technique	Optimism in the face of uncertainty	

N. Srinivas et al. "Gaussian process optimization in the bandit setting: No regret and experimental design." ICML 2010. (The 1st presentation today!)

Related Work: Maximum Entropy RL

What about collecting data by maximum entropy or entropy-regularized RL?

$$\max_{\pi} \mathrm{H}\left(\mathrm{d}_{\mu}^{\pi}\right), \quad \max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^{t}(r(s_t, a_t) - \beta \log \pi(a_t|s_t)) \,\middle|\, s_0 \sim \mu, \pi\right].$$

- No guarantee for arbitrary reward functions in downstream tasks.

E. Hazan et al. "Provably efficient maximum entropy exploration." ICML 2019. (The 3rd presentation today!)

B. Eysenbach and S. Levine. "Maximum entropy RL (provably) solves some robust RL problems." ICLR 2022.

Related Work: Reward-Free RL

- Study this problem in tabular settings or impose structural assumptions on values function and transitions.
- No practical algorithms.

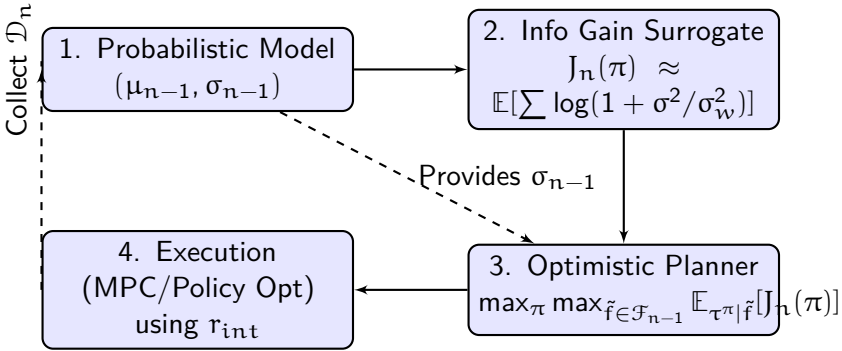
C. Jin et al. "Reward-free exploration for reinforcement learning." ICML 2020.
R. Wang et al. "On reward-free reinforcement learning with linear function approximation." NeurIPS 2020.
J. Chen et al. "On the statistical efficiency of reward-free exploration in non-linear RL." NeurIPS 2022.

Contributions

- Propose OpAX, a practical algorithm for learning non-linear dynamics with continuous state-action spaces and without structural assumptions.
- In this setting, OpAX is the first algorithm to have a theoretical guarantee for a rich family of nonlinear dynamics.

Outline

OPAX Components



- OPAX combines these components in an episodic loop.

Component 1: Probabilistic Dynamics Model

- Needed to represent belief about f^* and quantify uncertainty.
- Provides mean prediction $\mu_{n-1}(x, u)$ and epistemic uncertainty $\sigma_{n-1}(x, u)$.
- **Examples:**
  - **Gaussian Processes (GPs):** Bayesian non-parametric. Posterior mean μ and variance σ^2. Strong theory, scalability challenges.
  - **Probabilistic Ensembles (PEs):** Ensemble of M NNs. Mean = average prediction, Uncertainty σ = disagreement (std dev). Scalable, heuristic uncertainty.
- **Crucial Requirement:** Model must provide *well-calibrated* uncertainty estimates.

$$\sigma_{n-1}(x, u) \text{ should reflect the true error } |\mu_{n-1}(x, u) - f^*(x, u)|$$

(Formalized in Assumption 1 of the paper).

Component 2: Information Gain Surrogate

- Direct computation of mutual information $I(f^*_{\tau^\pi}; y_{\tau^\pi} | \mathcal{D}_{1:n-1})$ is hard.
- **Approximation:** Relate information gain to reduction in predictive variance (epistemic uncertainty). For Gaussian noise $w_k \sim \mathcal{N}(0, \sigma_w^2 I)$, gain at one point (x, u) is approx:

$$I(f^*(x,u); y | \mathcal{D}_{n-1}) \approx \frac{1}{2} \sum_{j=1}^{d_x} \log \left(1 + \frac{\sigma^2_{n-1,j}(x,u)}{\sigma_w^2} \right)$$

- **OPAX Planning Objective (Intrinsic Reward):** Maximize expected cumulative log uncertainty reduction over the trajectory:

$$J_n(\pi) = \mathbb{E}_{\tau^\pi} \left[\sum_{t=0}^{T-1} \underbrace{\sum_{j=1}^{d_x} \frac{1}{2} \log \left(1 + \sigma_w^{-2} \sigma^2_{n-1,j}(x_t, \pi(x_t)) \right)}_{r_{int}(x_t, \pi(x_t))} \right] \quad \text{(Eq. 6)}$$

Component 3: Optimistic Planning

- **Problem:** The expectation $\mathbb{E}_{\tau^\pi}[\cdot]$ in Eq. 6 depends on the *unknown* true dynamics f^*. Planning with the mean model μ_{n-1} can be suboptimal (might avoid truly informative regions).
- **OPAX Solution: Optimism in Information Space**
  - Define a plausible set of dynamics models $\mathcal{F}_{n-1}$ based on current uncertainty σ_{n-1}.

$$\mathcal{F}_{n-1} = \{\tilde{f} \mid \|\tilde{f}(z) - \mu_{n-1}(z)\| \leqslant \beta\sigma_{n-1}(z) \forall z\} \quad \text{(Simplified concept)}$$

  - Plan to maximize the intrinsic reward objective (Eq. 6) under the *most favorable* dynamics $\tilde{f} \in \mathcal{F}_{n-1}$ for achieving high information gain:

$$\max_{\pi \in \Pi} \max_{\tilde{f} \in \mathcal{F}_{n-1}} \mathbb{E}_{\tau^\pi|\tilde{f}} \left[\sum_{t=0}^{T-1} r_{int}(x_t, \pi(x_t)) \right] \quad \text{(Eq. 7)}$$

- **Intuition:** Act as if the world works in the way (among plausible options) that best allows reaching uncertain, informative states. Encourages exploration.

Component 4: Execution Loop

- The optimistic planning problem (Eq. 7) is an optimal control problem with intrinsic reward r_{int} and optimistic dynamics $\tilde{f}$.
- **Solvers:**
  - **Model Predictive Control (MPC):** Replan at each step k using current state x_k, optimistic dynamics $\tilde{f}$, and r_{int}. Execute first action. (e.g., using iCEM).
  - **Policy Optimization:** Train policy π_θ (e.g., NN) using model-based RL (e.g., SAC) with rollouts from $\tilde{f}$ and reward r_{int}.
- **Overall OPAX Loop:**

```
for episode n = 1 to N do
    Compute optimistic policy π_n by solving Eq. 7 (using MPC or Policy Opt).
    Execute π_n for T steps, collect trajectory data D_n = {(x_t, u_t, x_{t+1})}_{t=0}^{T-1}.
    Update probabilistic model (μ_n, σ_n) using all data D_{1:n}.
```

Outline

Assumptions

Regularity of the Dynamics

$f^\star$ lies in a RKHS with kernel k and $\|f^\star\| \leqslant 1$.

Recall

$$\gamma_n(k) = \max_{\mathcal{D}_1,\ldots,\mathcal{D}_n} \log\det(I + K_n)$$

and

$$\gamma_n(\text{linear}), \gamma_n(\text{RBF}) \asymp \text{polylog}(nT).$$

N. Srinivas et al. "Gaussian process optimization in the bandit setting: No regret and experimental design." ICML 2010. (The 1st presentation today!)

Main Guarantee

Theorem (Informal)

If we use $\mathcal{GP}(0, k)$ to build well-calibrated confidence intervals, then w.h.p.

$$E_\pi\left[\underbrace{\max_t \|\sigma_n(s_t, a_t)\|_2^2}_{\substack{\textit{maximum uncertainty} \\ \textit{along the trajectory}}}\right] \leqslant \textit{poly}(T) \cdot \frac{\gamma_n(k)}{\sqrt{n}}, \quad \forall \pi,\ \forall n \in \mathbb{N}.$$

For linear and RBF kernels,

$$\lim_{n\to\infty} \sigma_n(s, a) \to 0.$$

Zero-Shot Performance

Corollary (Informal)

Consider an RL problem with a known bounded reward:

$$\arg\max_{\pi} R(\pi), \quad s_{t+1} = f^{\star}(s_t, a_t) + w_t.$$

If $n \gtrsim \mathrm{poly}(d, T, \varepsilon^{-1}, \gamma_n(k))$, *then*

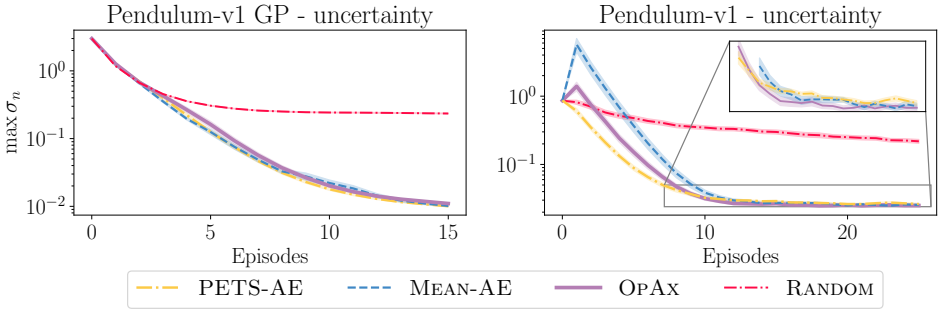
$$\max_{\pi} R(\pi) - R(\hat{\pi}) \leqslant \varepsilon,$$

where

$$\hat{\pi} \in \arg\max_{\pi} \min_{\eta} R(\pi, \eta), \quad s_{t+1} = \mu_n(s_t, a_t) + \eta(s_n)\beta_n\sigma_n(s_t, a_t) + w_t.$$

Outline

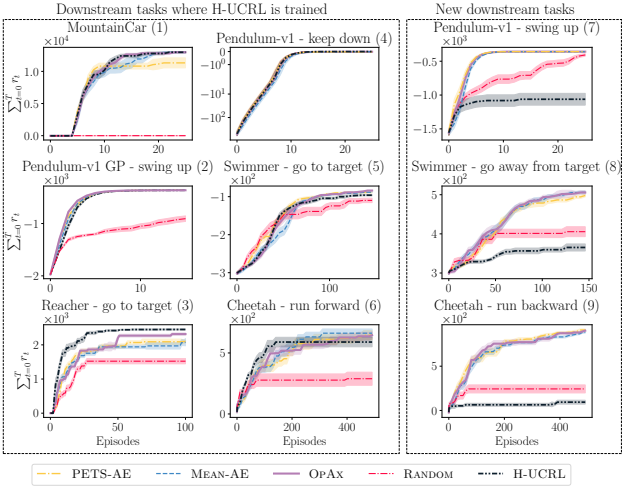
Experiments: Reduction in Epistemic Uncertainty (ok)



Active exploration reduces epistemic uncertainty faster than random exploration

Experiments: Learning and Zero-Shot Generalization (ok)

- compared to other active exploration baselines, `OpAx` constantly performs well and is **on par** with `H-UCRL`
- on the **unseen** tasks the active exploration baselines and `OpAx` **outperform** `H-UCRL` by a large margin



Experiments: Zero-Shot Pendulum Swing-Up (ok)

H-UCRL

OpAx

Key Findings

- **Sample Efficiency:**
  - OPAX often reaches target performance with fewer environment interactions.
- **Scalability:**
  - Successful application to high-dimensional Fetch task using PEs and MPC.
- **Objective Robustness:**
  - Log-uncertainty objective (Eq. 6) performs similarly to simpler sum-of-uncertainties (OPAX-SUM, Fig 5). Core benefit seems to be optimistic planning towards uncertainty.

Outline

Strengths and Strategic Relevance

- **Principled:** Grounded in information theory (Bayesian optimal experimental design).
- **Generalization Focus:** Directly tackles task-specific bias, aiming for versatile world models.
- **Theoretical Support:** Convergence guarantees (esp. for GPs) provide confidence.
- **Novel Optimism:** Applies optimism principle effectively to information gain.
- **Empirically Validated:** Strong performance across diverse benchmarks.
- **Strategic Importance:** Contributes to foundational challenge of building agents that autonomously acquire comprehensive world models – crucial for AGI goals (planning, reasoning, adaptation). OPAX offers a concrete algorithm for the data acquisition phase.

Limitations and Open Questions

- **Computational Cost:** Optimistic planning (solving Eq. 7) can be expensive, especially for high dimensions / long horizons. Scalability remains a challenge.
- **Uncertainty Calibration:** Performance heavily relies on accurate, well-calibrated uncertainty estimates from the probabilistic model. What if uncertainty is poor? (Robustness needed).
- **Assumptions:**
  - Fully Observed Systems (MDPs): How to extend to POMDPs?
  - Static Dynamics: How to handle non-stationarity?
  - Horizon Dependence: Theoretical bounds can depend on T.
- **Exploration vs. Exploitation:** OPAX is pure exploration. How to best integrate with downstream task execution?

Future Directions

- **Scalability & Efficiency:** Faster optimistic planning approximations (e.g., amortized planning).
- **Robustness:** Design methods less sensitive to uncertainty miscalibration.
- **POMDP Extension:** Information gain in belief space.
- **Advanced Models:** Apply to hierarchical / latent variable models.
- **Lifelong Learning:** Integrate OPAX into agents that learn continuously.
- **Balancing Objectives:** Combine information gain with task rewards during exploration.
- **Real-World Robotics:** Address safety, sample efficiency, physical interaction challenges.

Conclusion

- OPAX[2] presents a principled and effective algorithm for **task-agnostic active exploration** in dynamical systems.
- By maximizing **information gain** via **optimistic planning**, it learns globally accurate models that enable strong **zero-shot generalization**.
- Supported by **theoretical guarantees** and validated by **empirical results** across diverse environments.
- Represents a significant step towards building more **general, adaptable, and data-efficient** learning agents.
- Key challenges remain in **scalability** and **robustness**, offering exciting avenues for future research.

[2] **sukhija2023optimistic**.

References I

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